



Thursday Feb 19

Theory of Conjugate dist.

Problem

A rare sickness effects 1/1000 inhabitants.

test for people with sickness $\rightarrow$ 99% positive

test for people without sickness $\rightarrow$ 2% false positive

If the result on test is positive, what is the $P(\text{of}^{\text{not}} \text{having the disease})$

AKA find: $P(\overline{\text{having}}|+)$

$$P(\overline{\text{having}}|+) = \frac{P(+|\overline{\text{having}}) \cdot P(\overline{\text{having}})}{P(+)}$$

$$P(\overline{\text{having}}) = 1 - 1/1000$$

$$P(+|\overline{\text{having}}) = 0.02$$

$$P(+)= P(+|\overline{\text{having}}) \cdot P(\overline{\text{having}}) + P(+|\text{having}) \cdot P(\text{having})$$

$$0.02 \cdot (1 - 0.001) + 0.99 \cdot 0.001$$

$$P(\overline{\text{having}}|+) = \frac{0.02 \cdot (.999)}{0.02 \cdot (.999) + 0.99 \cdot 0.001} = 0.9527 //$$

Problem

let $X|\lambda \sim \text{Exponential}(\lambda)$ and $\lambda \sim \text{Gamma}(\alpha, \beta)$

a) obtain the joint distribution of (X, λ)

we know $X|\lambda \sim f(x|\lambda) = \lambda \exp(-\lambda x)$ and $\lambda \sim f(\lambda) = \frac{\beta^\alpha}{\Gamma(\alpha)} \lambda^{\alpha-1} \exp(-\lambda \beta)$

$$\text{So, } f(x, \lambda) = f(x|\lambda) \cdot f(\lambda) = \frac{\beta^\alpha}{\Gamma(\alpha)} \lambda^\alpha e^{-\lambda(\beta+x)}$$

b) marginal Expectation of X

$$E[X] = E[E[X|\lambda]] = E[1/\lambda] = \frac{\beta}{\alpha-1}$$

$$E[1/\lambda] = \int_0^\infty \frac{1}{\lambda} \frac{\beta^\alpha}{\Gamma(\alpha)} \lambda^{\alpha-1} e^{-\lambda \beta} d\lambda = \frac{\beta^\alpha}{\Gamma(\alpha)} \int_0^\infty \lambda^{\alpha-2} e^{-\lambda \beta} d\lambda = \frac{\beta^\alpha}{\Gamma(\alpha)} \int_0^\infty \lambda^{(\alpha-1)-1} e^{-\lambda \beta} d\lambda = \text{Gamma}(\alpha-1, \beta) \rightarrow \frac{\Gamma(\alpha-1)}{\beta^{\alpha-1}} \rightarrow$$

$$E[X] = \frac{\beta^\alpha}{\Gamma(\alpha)} \cdot \frac{\Gamma(\alpha-1)}{\beta^{\alpha-1}} = \frac{\beta(\alpha-1)}{(\alpha-1)\Gamma(\alpha-1)} = \frac{\beta}{\alpha-1} //$$

c) obtain the marginal distribution of X

Find $f(x)$

$$f(x) = \int_0^{\infty} f(x|\lambda) f(\lambda) d\lambda = \int_0^{\infty} \frac{\beta^a}{\Gamma(a)} \lambda^a \exp(-\lambda(\beta+x)) d\lambda = \frac{\beta^a}{\Gamma(a)} \cdot \frac{\Gamma(a+1)}{(\beta+x)^{a+1}} = \frac{\beta^a \Gamma(a)}{\Gamma(a)(\beta+x)^{a+1}} = \frac{\beta^a}{(x+\beta)^{a+1}}$$

get here by comparing to Normal Kernel $(\lambda)^{\text{shape}-2} e^{-(\text{rate})\lambda}$
 $\text{shape} = a+1$
 $\text{rate} = \beta+x$
 $\Gamma(a+1) = a\Gamma(a)$
 Finding Integral value
 $\int_0^{\infty} \lambda^a e^{-\lambda(\beta+x)} d\lambda = \frac{\Gamma(a+1)}{(\beta+x)^{a+1}}$

d) Conditional distribution of λ given $X=x$

$$f(\lambda|x) \propto f(x|\lambda) \cdot f(\lambda) = \text{gamma}(a+1, \beta+x)$$

Problem

Let $X|\mu \sim \text{Normal}(\mu, 1)$ and $\mu \sim \text{Normal}(0, 1)$

a) obtain joint distribution of (X, μ)

$$f(x|\mu) = \frac{1}{\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2}} \quad \text{and} \quad f(\mu) = \frac{1}{\sqrt{2\pi}} e^{-\frac{\mu^2}{2}}$$

$$f(x, \mu) = f(x|\mu) \cdot f(\mu) = \frac{1}{\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2}} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{\mu^2}{2}} = \frac{1}{2\pi} e^{-\frac{(x^2 - 2\mu x + \mu^2 + \mu^2)}{2}} //$$

$(x-\mu)^2 \rightarrow x^2 - 2\mu x + \mu^2 + \mu^2 = x^2 + 2\mu^2 - 2\mu x$

b) obtain Marginal Expectation and Variance of X

$$E[X] = E[E[X|\mu]] = E[\mu] = 0$$

$$V[X] = E[V[X|\mu]] + V[E[X|\mu]] = E[1] + V[\mu] = 1 + 1 = 2$$

c) obtain marginal distribution of X

$$\int \frac{1}{2\pi} \exp\left(-\frac{x^2 + 2\mu^2 - 2\mu x}{2}\right) d\mu = \text{Normal}(0, 2)$$

d) Conditional of μ given $X=x$

$$f(\mu|x) \propto f(x|\mu) \cdot f(\mu) = \frac{1}{2\pi} e^{-\frac{(x^2 + 2\mu^2 - 2\mu x)}{2}} //$$

Thursday 19th
Worksheet 2